

Valuation accuracy: measured against Collectr

Today's work addressed six defects raised from client testing. The system no longer grades cards; it identifies the printing, reads the grading label, and prices that exact grade from completed sales. Accuracy is stated against Collectr as the reference, on the cards where a direct comparison exists.

13 commits 55 automated tests, all passing 24 new slab fixtures 14 grading companies recognised

Accuracy against the reference

CARD	GRADE READ	COLLECTR	GRAILCARD NOW	VARIANCE	BEFORE TODAY
Deoxys ex · EX Emerald #93	PSA 9	\$1,864.96	\$1,868.84	+0.21%	\$903.24
Charizard ☆ δ Gold Star · EX Dragon Frontiers #100	BGS 8.5	\$15,000.00	\$14,299.00	-4.7%	\$15.56
Umbreon VMAX alt-art · Evolving Skies #215	PSA 10	—	\$4,250.00	no reference	\$200 (wrong card #95)
Charizard · Legendary Collection #3	PSA 5	—	\$529.59 (PSA 8 rung)	no reference	wrong card entirely

The Gold Star variance is expected and disclosed in-product: it is a **Beckett** slab and every sale comp available to us is a **PSA** sale, so the figure is the nearest PSA tier rather than a recorded BGS sale. The interface says so rather than presenting it as a match.

Issues raised by the client, and what each turned out to be

REPORTED	ACTUAL CAUSE	RESOLUTION
Wrong card identified	The grading label was read but discarded. Where the label's grade digit was unreadable, the whole label was thrown away, dropping the card to fuzzy name matching — a Legendary Collection Charizard resolved to Dragon Frontiers, and a Deoxys ex to POP Series 4.	Failing to read a grade is no longer treated as evidence the card is ungraded. The label's set and number stay available for identification, and every plausible label line is scored against the catalogue.
Beckett shown as PSA	When the company logo failed to OCR, the code <i>guessed</i> , and the guess defaulted to PSA.	Detection now keys on markers Beckett prints and PSA never does — the four subgrade captions and the “+” suffix. Where nothing indicates a company it reports unknown instead of inventing one.
Grading wildly wrong 51 marks on a clean card	The v0 heuristics were not fit to move money, and their output was also being applied as a price multiplier.	Grading removed entirely — see below.
PSA 9 priced at ~\$150	Two faults stacked: the wrong card, and a raw price quoted for a card known to be slabbed.	A raw price is never quoted for a slabbed card. Where the grade cannot be read the price is withheld and the state is named.
Raw card under-valued \$84 shown as \$21	Our own grade of 2.5 was applied as a 0.25 multiplier against the market price.	Condition adjustment removed with the grade that produced it. A raw card is quoted at the raw market price.
Prices showing \$NaN	Deploy skew — the browser held an older bundle expecting a plain number while the API had moved to a richer object.	Tiles accept either shape. A stale price is recoverable; a price reading NaN is not.

The two changes that produced most of the accuracy gain

Reading the provider's filtered price

The price source publishes two figures per grade: a raw median and a filtered, recency-weighted price. We were reading the raw one. On the Deoxys, the sample ran from \$80.63 to \$1,882.55 for the same card at the same grade — mistitled and damaged listings sitting in the median at full weight. The filtered figure is \$1,868.84. Across the tested set the two differ by **~51% to +107%**.

Reading the grade word, not the digit

PSA prints the same fact twice, and the two are not equally legible. OCR read the isolated 9 on a MINT label as a 6, and sometimes dropped it entirely. PSA's scale is published and fixed — MINT is always 9 — so the multi-character word now decides and the digit only confirms.

Removed

- Our **condition grading**, on every card — slabbed and raw
- Sub-grades** for corners, edges and surface
- Surface-mark detection** and the annotated detection view
- Centering measurement** and its PSA pass/fail verdict
- The condition multiplier** that discounted market prices
- Grade / don't-grade recommendations** and net-profit tables
- Provisional grades** previously returned on rejected photos

A card already certified by PSA or Beckett does not need our opinion, and for a raw card the heuristics were not accurate enough to influence a price.

Added

- A **grade tuple** — company, grade, qualifier, label variant, sub-grades, tier — replacing a single number
- 14 grading companies**, including Beckett half-grades, BVG, BCCG, Black Label, CGC's two tens and SGC's legacy 100-point scale
- Guards** against “PSA 10 candidate”, multi-card lots and Raw Card Review being read as certifications
- Prices per grading company**, with an honest empty state where we hold no data for that company
- Sample size and confidence** on every figure; a single-sale price is flagged as an anecdote
- Rejection of unreadable slab photos** rather than guessing
- Paste-a-screenshot** capture

Label-reading accuracy, measured

21%

slab fixtures passing before today (5/24)

100%

passing now (24/24)

4/4

real client photos identified and graded correctly

55

automated tests, all passing

The fixture set encodes the failure catalogue: Beckett 9.5 with sub-grades, Black Label, BCCG (a discount tier that a naive parser prices as a gem 10 — a 10–50× overvaluation), BVG, Raw Card Review, PSA qualifiers, CGC's ambiguous 10, SGC 88 on the legacy scale, and decimal grades from emerging graders.

Figures are estimates of market value drawn from completed sales, not appraisals. Comparison prices observed on Collectr, 24 August 2026. Beckett-certified cards are priced from the nearest PSA tier because no Beckett sales feed is currently available; this is disclosed in-product on every affected card.